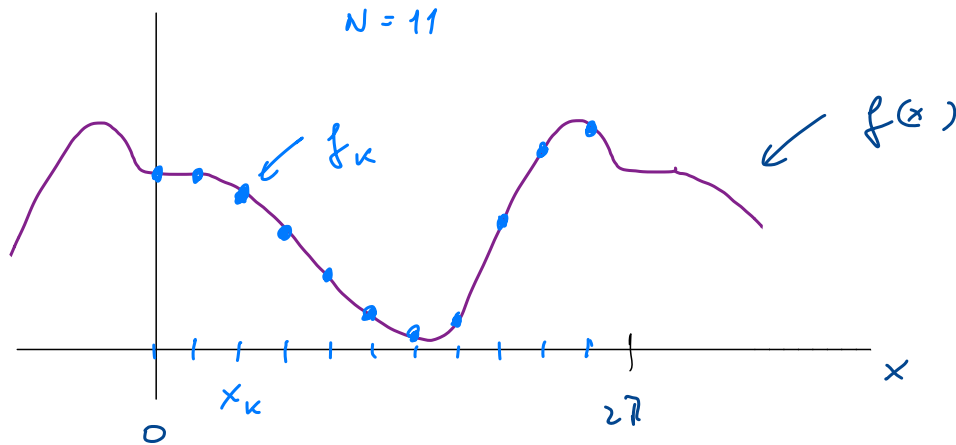


Discrete Fourier Transformation

$f(x)$ 2π -periodic

sample at N points: $x_k = \frac{2\pi}{N} \cdot k \quad k=0, \dots, N-1$

f takes values $f_k \equiv f(x_k) = f\left(\frac{2\pi}{N} \cdot k\right)$



We know: $f(x) = \sum_{n=-\infty}^{\infty} c_n e^{i n \cdot x}$

Q: $c_n = ?$

observation: c_n and c_{n+N} give the same values at x_k :

$$e^{i(n+N)x_k} = e^{i n \cdot x_k} \underbrace{e^{i N \frac{2\pi}{N} \cdot k}}_{e^{i 2\pi \cdot k} = 1} = e^{i n \cdot x_k}$$

$\Rightarrow$ we keep $n = 0, 1, \dots, N-1$, and determine $c_0, c_1, \dots, c_{N-1}$ such that

$$f_k \equiv \sum_{n=0}^{N-1} c_n e^{i n \cdot x_k}$$

trigonometric
interpolation

$$f_k = \sum_{n=0}^{N-1} c_n e^{i n \cdot \frac{2\pi}{N} \cdot k}$$

1. multiply by $e^{-i m \cdot X_k} = e^{-i \frac{2\pi}{N} \cdot k \cdot m}$
2. Sum over k

$$\text{l.h.s.} \quad \sum_{k=0}^{N-1} f_k \cdot e^{-i \frac{2\pi}{N} \cdot k \cdot m}$$

$$\text{r.h.s.} \quad \sum_{k=0}^{N-1} \sum_{n=0}^{N-1} c_n \cdot \underbrace{e^{i n \cdot \frac{2\pi}{N} \cdot k} e^{-i m \cdot \frac{2\pi}{N} \cdot k}}_{e^{i \frac{2\pi}{N} (n-m) \cdot k} = \left(e^{i \frac{2\pi}{N} (n-m)} \right)^k}$$

$r = r(n-m)$

exchange
sums

$$\downarrow$$

$$= \sum_{n=0}^{N-1} c_n \sum_{k=0}^{N-1} r^k$$

$$- \text{ if } n=m \Rightarrow r=1 \Rightarrow \sum_{k=0}^{N-1} 1^k = N$$

- if $n \neq m$

$$\sum_{k=0}^{N-1} r^k = \frac{r^N - 1}{r - 1} \underset{\uparrow}{=} \frac{1 - 1}{r - 1} = 0 \quad \text{since}$$

$$r^N = \left(e^{i \frac{2\pi}{N} (n-m)} \right)^N = e^{i 2\pi (n-m)} = 1$$

$$\Rightarrow \sum_{k=0}^{N-1} r^k = N \cdot \delta_{n,m}$$

$$\text{r.h.s.} \quad \sum_{n=0}^{N-1} c_n \cdot N \cdot \delta_{n,m} = N \cdot c_m$$

$$\Rightarrow \sum_{k=0}^{N-1} f_k e^{-i \frac{2\pi}{N} \cdot k \cdot m} = N \cdot c_m$$

replace m by n $\Rightarrow$

fitted sampled values

Fourier coefficients $C_n = \frac{1}{N} \sum_{k=0}^{N-1} f_k e^{-i \frac{2\pi}{N} k \cdot n}$ ✓

Def. $\hat{f}_n \equiv \sum_{k=0}^{N-1} e^{-i \frac{2\pi}{N} k \cdot n} f_k$ (1)

↑
discrete F.T. of $f(x)$

$f(x_k) = \frac{1}{N} \sum_{n=0}^{N-1} \hat{f}_n e^{i \frac{2\pi}{N} k \cdot n}$ (2)

Remark:

• this is a matrix multiplication!

$w \equiv e^{-i \frac{2\pi}{N}}$, $F_{nk} \equiv (w)^{k \cdot n}$

$$\begin{bmatrix} \hat{f}_0 \\ \vdots \\ \hat{f}_{N-1} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & \dots \\ 1 & w & w^2 & \\ & 1 & w & w^2 \\ 1 & w^2 & w^4 & \\ \vdots & & & \ddots \end{bmatrix} \cdot \begin{bmatrix} f_0 \\ \vdots \\ f_{N-1} \end{bmatrix} \Rightarrow \underline{\hat{f}} = \underline{F} \cdot \underline{f}$$

$\hat{f}$ $\underline{F}$ f

(2) can be rewritten as

$\underline{f} = \frac{1}{N} \underline{F}^* \cdot \underline{\hat{f}} = \frac{1}{N} \underbrace{\underline{F}^* \cdot \underline{F}}_{\text{must be } \underline{I}} \cdot \underline{f} \Rightarrow \underline{F}^{-1} = \frac{1}{N} \underline{F}^*$

$\left(\underline{U} \equiv \frac{1}{\sqrt{N}} \underline{F} \text{ unitary} \right)$

• $N = 2^M \rightarrow \exists \text{ FFT } \sim$
Fast Fourier Transformation

run time $\sim N \ln N$

• Discrete F.T. as integral approximation !

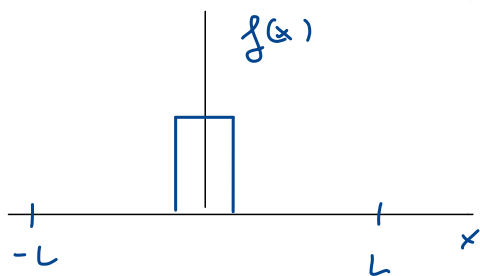
$$\sqrt{2\pi} \cdot \hat{f}(\omega) \approx \int_{-L}^L dx e^{-i\omega \cdot x} f(x) \approx \frac{2L}{N} \sum_{x_k} e^{-i\omega \cdot x_k} f_k$$

Δx

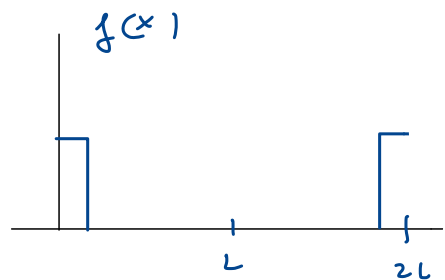
set $\omega \rightarrow \omega_n = \frac{\pi}{L} \cdot n \quad n=0, \dots, N-1 \Leftarrow x_k = \frac{2L}{N} \cdot k$

$$\hat{f}(\omega = \frac{\pi}{L} \cdot n) \approx \left(\frac{2L}{\sqrt{2\pi} \cdot N} \right) \cdot \sum_{k=0}^{N-1} e^{-i \frac{2\pi}{N} \cdot n \cdot k} \cdot f(x_k)$$

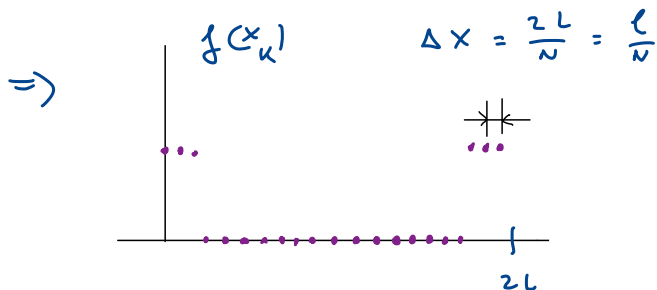
In practice:



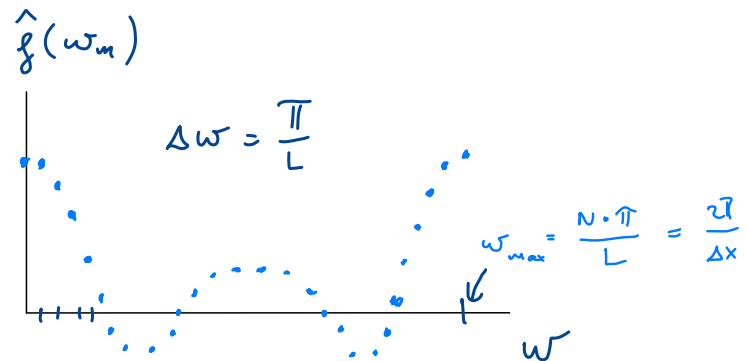
fold
back
 $\Rightarrow$



discretize

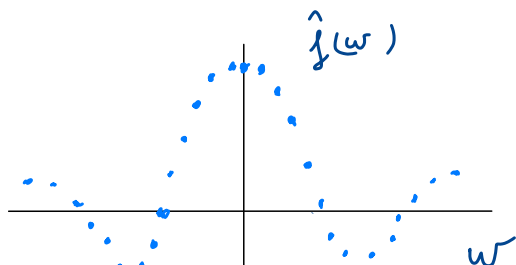


$\Rightarrow$
DFT

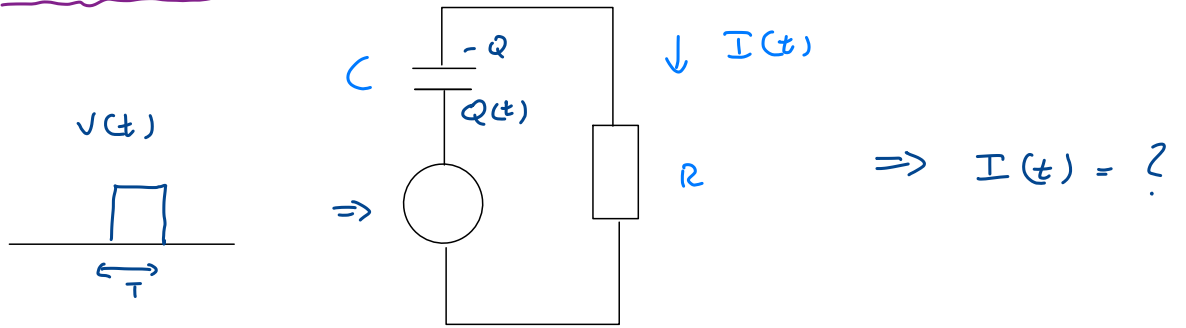


fold
back

$\Rightarrow$



Simple application



simplification (first)

$$V(t) = \frac{Q(t)}{C} + I(t) \cdot R$$

$$\frac{d}{dt} \Rightarrow \dot{V}(t) = \frac{1}{C} I(t) + R \cdot \dot{I}(t) \quad (*)$$

if they are equal $\rightarrow$ their Fourier transforms are also!

$$V(t) \leftrightarrow \hat{V}(\omega)$$

$$\dot{V}(t) \leftrightarrow i\omega \hat{V}(\omega)$$

$$I(t) \leftrightarrow \hat{I}(\omega)$$

$$\dot{I}(t) \leftrightarrow i\omega \cdot \hat{I}(\omega)$$

$$i\omega \hat{V}(\omega) = \frac{1}{C} \hat{I}(\omega) + i\omega \cdot R \hat{I}(\omega)$$

$$\hat{V}(\omega) = \underbrace{\left(\frac{1}{i\omega \cdot C} + R \right)}_{Z(\omega)} \hat{I}(\omega)$$

$Z(\omega)$ impedance!

$|Z(\omega)|$ determines size of signal!

its phase $\sim$ delay

A.C. voltage :

$$V(t) = V_0 \cos(\omega t) \rightarrow \frac{V_0}{2} e^{i\omega t} + \text{c.c.}$$

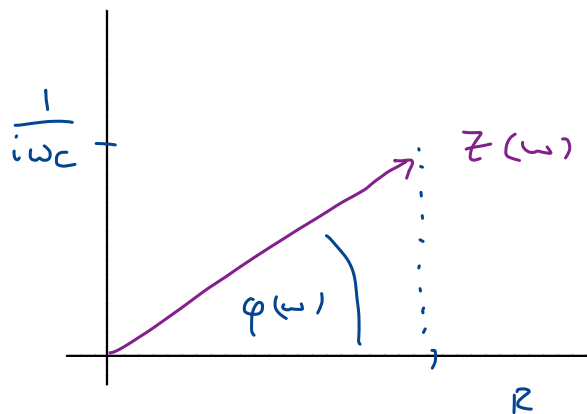
$$\hat{V}(\omega) = \frac{V_0}{2}$$

$$Z(\omega) = |Z(\omega)| \cdot e^{i\varphi(\omega)} \Rightarrow \hat{I}(\omega) = \frac{V_0}{2} |Z| e^{i\varphi}$$

$$\Rightarrow I(t) = \frac{V_0}{2} |Z(\omega)| e^{i\varphi} e^{i\omega t} + \text{c.c.} =$$

$$Z(-\omega) = Z(\omega)^* = |Z| \cdot e^{-i\varphi}$$

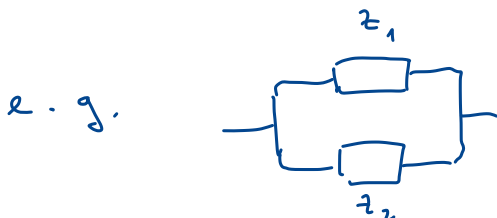
$$I(t) = V_0 |Z(\omega)| \cos(\omega t + \varphi(\omega))$$



• for coils

$$V = L \cdot \dot{I} \Rightarrow Z = i\omega \cdot L$$

• usual relations carry over !



$$\Rightarrow Z = \frac{Z_1 \cdot Z_2}{Z_1 + Z_2}$$